

Project Management Tools

COMS W4156: Advanced Software Engineering
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About Me

- Undergrad at Columbia — SEAS CS '23
- Returned through MS Express
 - IS/ML tracks
 - 4 SWE internships
 - 6th semester TAing
- You've probably seen me on Ed (and [r/columbia](#))



Agenda

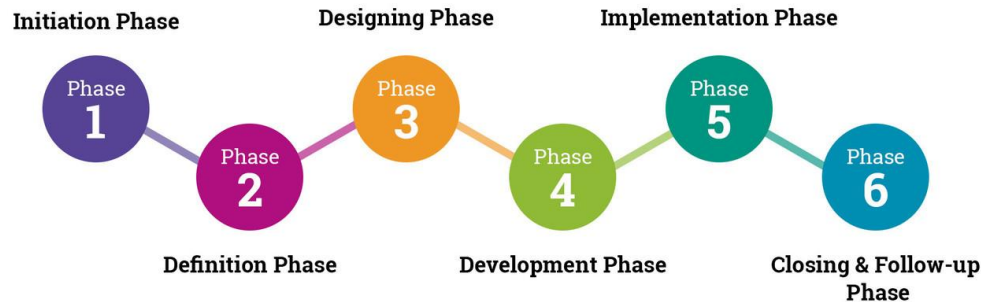
1. Intro
2. Jira
3. Trello
4. Which One Should You Use?
5. Demo
6. Best Practices

Intro

Intro

- Organize tasks and deadlines
- Track project progress
- Improve team collaboration
- Ensure accountability

Project Management Process Phases





Jira

Jira - Overview

- Created by Atlassian, popular for agile project management
- Primarily used for software development
- Supports features like sprint planning, issue tracking, and roadmaps



Jira - Key Features

- Boards: Visualize tasks using Kanban or Scrum boards
- Issues: Track bugs, tasks, and user stories
- Sprints: Break work into 1–4 week cycles
- Backlog: Prioritize and plan upcoming work
- Reports: Track progress using burndown charts, velocity charts, etc



Board



Epic ▾

GROUP BY

Choices ▾

TO DO 12

Implement feedback collector

NUC-205

9



Bump version for new API for billing

NUC-206

3



Add NPS feedback to wallboard

NUC-208

1



Add analytics events to pricing page

NUC-209

3



Resize the images for the upcoming campaign

NUC-210

1



IN PROGRESS 4

Update T&C copy with v1.9 from the writers guild in all products that have cross country compliance

NUC-213



1



Bump feedback icon version

NUC-214

3



Tech spike on new stripe integration with paypal

NUC-215

3



Change phone number field type to 'phone'

NUC-217



1



IN QA 4

Adapt web app no new payments provider

NUC-346



5



Purchasing error - edit fields

NUC-354



3



Multi-dest search UI web

NUC-338

5



DONE 4

Quick booking for accomodations - web

NUC-336



4



Fluid booking on tablets

NUC-343



5



Shoping cart purchasing error - quick fix required.

NUC-354



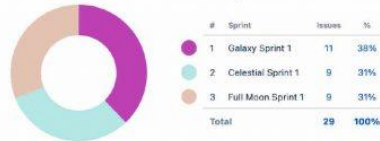
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Program Level Dashboard

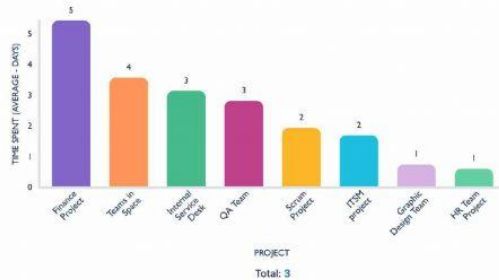
Custom Charts

Issues in Active Sprints



Custom Charts

Average Time Spent on Work by Project



Projectrak Two Dimensional Project Statistics

	ON TRACK	AT RISK	PAUSED	PENDING	Projects per value
Graphic Design Team	1	0	0	0	1
HR Team Project	0	0	1	0	1
Internal Service Desk	1	0	0	0	1
ITSM project	0	0	0	1	1
QA Team	0	1	0	0	1
Scrum Project	0	1	0	0	1
Teams in Space	0	1	0	0	1
Finance Project	1	0	0	0	1
Projects per value	3	3	1	1	8

Values: 4 OF 4 STATUS / 8 OF 8 NAME

Custom Charts

Status of Issues Across Projects

Project	Status	To Do	In Progress	Blocked	Done	Total
Finance Project		1	3	0	6	16
Teams in Space		7	2	2	6	17
Internal Service Desk		1	7	-	4	12
QA Team		5	3	-	2	10
Scrum Project		23	5	-	7	35
Show more...		6	7	5	138	156
Total (Issues)		43	27	13	163	246

Custom Charts

Issues in Each Release



Projectrak Project List

Name	Category	Created date	Priority	Status
Finance Project	Business	28/Sep/22	Medium	ON TRACK
Graphic Design Team	Business	13/Dec/22	Medium	ON TRACK
HR Team Project	Business	12/Oct/22	Low	PAUSED
Internal Service Desk	Service Management	16/Nov/22	High	ON TRACK
ITSM project	Service Management	01/Dec/22	Medium	PENDING
QA Team	Software Development	14/Sep/22	Blocker	AT RISK
Scrum Project	Software Development	05/Jan/23	Blocker	AT RISK
Teams in Space	Software Development	27/Oct/22	Blocker	AT RISK

8 results

Projectrak Project Pie Chart





Trello

Trello - Overview

- Lightweight, visual project management tool owned by Atlassian since 2017
- Uses boards, lists, and cards
- Great for simpler project management or smaller teams



Trello - Key Features

- Boards: Organize tasks and workflows
- Lists: Divide tasks by stages (Backlog, In Progress, Done)
- Cards: Represent individual tasks or items
- Power-ups: Extend Trello's functionality (e.g. calendar view, custom fields)
- Collaboration: Commenting, file sharing, task assignments



Marketing Board

Backlog

- Email Campaign
Monthly Newsletter
- Email Campaign
Onboarding Campaign
- Blog
Blog: How to Run a Team Meeting Effectively
- Social Media
Promo Recording

+ Add a card

In Progress

- Blog
Blog: Keeping Focus While Working Remotely
- Email Campaign
Monthly Newsletter
- Website
Feature Page
- Website
Customer Feedback Page
- Social Media

+ Add a card

Review

- Blog SEO
Post: How to Organize Remote Meetings
- Email Campaign
Black Friday Banner

+ Add a card

Done

- Email Campaign
Trial Emails
- Email Campaign
Daunting Campaign
- Website
Website Redesign

+ Add a card

+ Add another list



Which One Should You Use?

Which One Should You Use?



- Designed for complex projects
- Supports agile methodologies
- Features like sprint planning and issue tracking
- Advanced reporting and analytics
- Generally higher cost for teams



- Ideal for simpler projects
- Uses a visual board system
- Flexible with lists, cards, and boards
- Basic features, easy to use
- Free tier available, with optional paid features



*“The wise ASE
student will save themselves
a hefty bill and
a headache with Trello.”*

—Sun Tzu, Art of War



Demo

Best Practices

Best Practices

- Update boards regularly
- Break down large tasks
- Use labels and due dates





Questions?